

Lecture 5

Friedman Test

Quick recap of lecture 4

1. Kruskal Wallis is used for more than two independent samples of either ordinal or non-normal quantitative data.
2. Tests the null hypothesis that the medians of the k populations are the same.
3. It achieves this by combining the samples, ranking their values and calculating the sum of ranks for each sample.
4. Must have 3 or more observations per group.
5. For larger sample sizes, the test statistic is approximately chi-square distributed with $k-1$ degrees of freedom.

Learning outcomes for this lecture

- 1) Know the conditions (data type, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the Friedman test
- 2) Know the data assumptions of the Friedman test
- 3) Be able to conduct the Friedman test

Friedman Test

- When is it appropriate? When you want to compare more than two groups/samples of ordinal or quantitative data, using matched or blocked samples, with respect to their locations (medians) i.e. determine whether they all come from the same population.
 - The test is an equivalent of randomised block design two-way ANOVA
- Data and Assumptions:
 - The data are either ordinal or quantitative but not normal.
 - The data are generated from a blocked experiment with b blocks
 - The measurements within a block are dependent/related.
 - The measurements from different blocks are independent.
 - No interaction between blocks and treatments.

Friedman Test – Randomized Block Design

Treatment	Blocks			
	1	2	3	4
1	y_{11}	y_{12}	y_{13}	y_{14}
2	y_{21}	y_{22}	y_{23}	y_{24}
3	y_{31}	y_{32}	y_{33}	y_{34}
4	y_{41}	y_{42}	y_{43}	y_{44}
5	y_{51}	y_{52}	y_{53}	y_{54}

"Treatments" are what are administered to the experimental units in an experiment. They refer to the populations/variables that are being compared in the test.

"Blocking" is introduced into an experiment in order to improve the comparison of the treatments by grouping experimental units into "blocks" (according to some characteristic) that are of the same size as the number of treatments that are being compared

Blocks (Experts)	A	B	C	D
E.B.	24	26	25	22
N.B.	27	27	26	24
M.D.	19	22	20	16
M.H.	24	27	25	23
B.J.	22	25	22	21
R.J.	26	27	24	24
B.K.	27	26	22	23
B.M.	25	27	24	21
J.S.	22	23	20	19

Friedman Test

- Hypotheses:

- H_0 : The locations of all the k populations (treatments) are the same
- H_1 : At least two population locations differ.
- Remember to INTERPRET your hypotheses from the information given to you in the question context!

- Calculating the Test statistic:

- 1) Rank the observations from smallest to largest within each block
- 2) Average ranks of tied observations within the same block
- 3) Calculate the rank sums $T_1, T_2, \dots, T_k$ for all the k treatments

Friedman Test

• Test statistic:

$$F_r = \left[\frac{12}{b(k)(k+1)} \sum_{j=1}^k T_j^2 \right] - 3b(k+1)$$

- where b is the number of blocks and
- k is the number of treatments
- The test statistic F_r has an approximate chi-squared distribution (provided that either k or $b \geq 5$) with degrees of freedom equal to k-1
- Critical region: We reject the null hypothesis if F_r is too large
i.e. Reject H_0 if $F_r \geq c$, where c is the critical value, OR the approximate/exact p-value $\leq \alpha$.

Friedman Test --- Example 1

Four managers evaluate applicants for a job in an accounting firm on several dimensions including academic credentials, previous work experience and personal suitability. Each manager then summarizes the results and produces an evaluation of the candidate. There are 5 possibilities:

- 1) The candidate is in the top 5% of applicants
- 2) The candidate is in the top 10% of applicants, but not in the top 5%
- 3) The candidate is in the top 25% of applicants, but not in the top 10%
- 4) The candidate is in the top 50% of applicants, but not in the top 25%
- 5) The candidate is in the bottom 50% of applicants

Eight applicants were randomly selected, and their evaluations by the four managers recorded. Can we conclude that there are differences in the way managers evaluate candidates?

Friedman Test --- Example 1

Applicant	Manager			
	1	2	3	4
1	2	1	2	2
2	4	2	3	2
3	2	2	2	3
4	3	1	3	2
5	3	2	3	5
6	2	2	3	4
7	4	1	5	5
8	3	2	5	3

Friedman Test --- Example 1

We are wanting to compare four dependent samples of ordinal data with respect to their locations, which are arranged in a block design use the Friedman Test

H_0 : There is no difference in the way that managers evaluate candidates

H_1 : There is a difference in the way that managers evaluate candidates

Calculating the test statistic:

$$k = 4$$

$$b = 8$$

Friedman Test --- Example 1

Ranking observations WITHIN BLOCKS:

Applicant	Manager			
	1	2	3	4
1	2 (3)	1 (1)	2 (3)	2 (3)
2	4	2	3	2
3	2	2	2	3
4	3	1	3	2
5	3	2	3	5
6	2	2	3	4
7	4	1	5	5
8	3	2	5	3

Friedman Test --- Example 1

Ranking observations WITHIN BLOCKS:

Applicant	Manager			
	1	2	3	4
1	2 (3)	1 (1)	2 (3)	2 (3)
2	4 (4)	2 (1.5)	3 (3)	2 (1.5)
3	2	2	2	3
4	3	1	3	2
5	3	2	3	5
6	2	2	3	4
7	4	1	5	5
8	3	2	5	3

Friedman Test --- Example 1

Ranking observations WITHIN BLOCKS:

Applicant	Manager			
	1	2	3	4
1	2 (3)	1 (1)	2 (3)	2 (3)
2	4 (4)	2 (1.5)	3 (3)	2 (1.5)
3	2 (2)	2 (2)	2 (2)	3 (4)
4	3 (3.5)	1 (1)	3 (3.5)	2 (2)
5	3 (2.5)	2 (1)	3 (2.5)	5 (4)
6	2 (1.5)	2 (1.5)	3 (3)	4 (4)
7	4 (2)	1 (1)	5 (3.5)	5 (3.5)
8	3 (2.5)	2 (1)	5 (4)	3 (2.5)

Friedman Test --- Example 1

Ranking observations WITHIN BLOCKS:

Applicant	Manager			
	1	2	3	4
1	2 (3)	1 (1)	2 (3)	2 (3)
2	4 (4)	2 (1.5)	3 (3)	2 (1.5)
3	2 (2)	2 (2)	2 (2)	3 (4)
4	3 (3.5)	1 (1)	3 (3.5)	2 (2)
5	3 (2.5)	2 (1)	3 (2.5)	5 (4)
6	2 (1.5)	2 (1.5)	3 (3)	4 (4)
7	4 (2)	1 (1)	5 (3.5)	5 (3.5)
8	3 (2.5)	2 (1)	5 (4)	3 (2.5)
SUM (ranks)	21	10	24.5	24.5
	T_1	T_2	T_3	T_4

Friedman Test --- Example 1

Test statistic:

$$\begin{aligned} F_r &= \left[\frac{12}{b(k)(k+1)} \sum_{j=1}^k T_j^2 \right] - 3b(k+1) \\ &= \left[\frac{12}{(8)(4)(5)} ((21)^2 + (10)^2 + (24.5)^2 + (24.5)^2) \right] - (3)(8)(5) \\ &= \frac{3}{40} (1741.5) - 120 \\ &= 10.61 \end{aligned}$$

Critical Region: $F_r \sim \chi^2_3$

Reject H_0 if $F_r \geq \chi^2_{3;0.05}$ OR if the approximate p-value ≤ 0.05

CHI-SQUARED DISTRIBUTION: One sided critical values, i.e. the value of $\chi_n^{2(P)}$ such that $P = \Pr[\chi_n^2 > \chi_n^{2(P)}]$, where n is the degrees of freedom, for $P < 0.5$

n	Probability Level P									
	0.4	0.2	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
1	0.708	1.642	2.706	3.841	5.024	6.635	7.879	9.140	10.827	12.115
2	1.833	3.219	4.605	5.991	7.378	9.210	10.597	11.983	13.815	15.201
3	2.946	4.642	6.251	7.815	9.348	11.345	12.838	14.320	16.266	17.731
4	4.045	5.989	7.779	9.488	11.143	13.277	14.860	16.424	18.466	19.998
5	5.132	7.289	9.236	11.070	12.832	15.086	16.750	18.385	20.515	22.106
6	6.211	8.558	10.645	12.592	14.449	16.812	18.548	20.249	22.457	24.102
7	7.283	9.803	12.017	14.067	16.013	18.475	20.278	22.040	24.321	26.018
8	8.351	11.030	13.362	15.507	17.535	20.090	21.955	23.774	26.124	27.867
9	9.414	12.242	14.684	16.919	19.023	21.666	23.589	25.463	27.877	29.667
10	10.473	13.442	15.987	18.307	20.483	23.209	25.188	27.112	29.588	31.419

Critical value = 7.815

p-value $\approx$ 0.0175

Friedman Test --- Example 1

Conclusion (a): Since $10.61 > 7.815$ we Reject H_0 and conclude that there are significant differences in the way managers evaluate candidates at the 5% significance level

OR

Conclusion (b): Since $p\text{-value} \approx 0.0175 (< 0.05)$ we Reject H_0 and conclude that there are significant differences in the way managers evaluate candidates at the 5% significance level

Friedman Test --- Example 2

- Four property development companies enter sealed bids for a number of vacant plots of land on auction. From the random sample of plots and bids (in thousands of Rands) made below, does it appear as if some firms tend to make higher bids on average than others?
- Test at the 2.5% significance level.

Plot	Company 1	Company 2	Company 3	Company 4
A	37	32	43	36
B	127	110	139	130
C	15	12	16	15
D	340	340	390	360
E	100	90	120	120
F	225	210	240	230

Friedman Test --- Example 2

We are wanting to compare four dependent samples of quantitative data with respect to their locations, which are arranged in a block design $\square$ use the Friedman Test

H_0 : There is no difference in the bids made by the four companies

H_1 : There is a difference in the bids made by the four companies

Calculating the test statistic:

$$k = 4$$

$$b = 6$$

Friedman Test --- Example 2

Ranking observations WITHIN BLOCKS:

Plot	Company 1	Company 2	Company 3	Company 4
A	37	32	43	36
B	127	110	139	130
C	15	12	16	15
D	340	340	390	360
E	100	90	120	120
F	225	210	240	230
SUM (ranks)				
	T_1	T_2	T_3	T_4

Friedman Test --- Example 2

Ranking observations WITHIN BLOCKS:

Plot	Company 1	Company 2	Company 3	Company 4
A	37 (3)	32 (1)	43 (4)	36 (2)
B	127 (2)	110 (1)	139 (4)	130 (3)
C	15 (2.5)	12 (1)	16 (4)	15 (2.5)
D	340 (1.5)	340 (1.5)	390 (4)	360 (3)
E	100 (2)	90 (1)	120 (3.5)	120 (3.5)
F	225 (2)	210 (1)	240 (4)	230 (3)
SUM (ranks)	13	6.5	23.5	17
	T_1	T_2	T_3	T_4

Friedman Test --- Example 2

Test Statistic:

$$F_r = \left[\frac{12}{b(k)(k+1)} \sum_{j=1}^k T_j^2 \right] - 3b(k+1)$$

$$= \left[\frac{12}{(6)(4)(5)} ((13)^2 + (6.5)^2 + (23.5)^2 + (17)^2) \right] - (3)(6)(5)$$

$$= \frac{3}{30} (1052.5) - 90$$

$$= 15.25$$

Critical Region:

$$F_r \sim \chi^2_3$$

Reject H_0 if

OR if the approximate p-value ≤ 0.025

$$F_r \geq \chi^2_{3;0.025}$$

CHI-SQUARED DISTRIBUTION: One sided critical values, i.e. the value of $\chi_n^{2(P)}$ such that $P = \Pr[\chi_n^2 > \chi_n^{2(P)}]$, where n is the degrees of freedom, for $P < 0.5$

n	Probability Level P									
	0.4	0.2	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
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7	7.283	9.803	12.017	14.067	16.013	18.475	20.278	22.040	24.321	26.018
8	8.351	11.030	13.362	15.507	17.535	20.090	21.955	23.774	26.124	27.867
9	9.414	12.242	14.684	16.919	19.023	21.666	23.589	25.463	27.877	29.667
10	10.473	13.442	15.987	18.307	20.483	23.209	25.188	27.112	29.588	31.419

Critical value = 9.348

p-value $\approx$ 0.00175

Friedman Test --- Example 2

Conclusion (a): Since $15.25 > 9.348$ we Reject H_0 and conclude that there is a significant difference in the bids made by the four companies at the 2.5% significance level i.e. some companies do make higher bids than others

OR

Conclusion (b): Since p-value $0.00175 < 0.025$ we Reject H_0 and conclude that there is a significant difference in the bids made by the four companies at the 2.5% significance level i.e. some companies do make higher bids than others

Q: What if the ranks were evenly spread? i.e. $T_1 = T_2 = T_3 = T_4 = 15$

$$F_r = \left[\frac{12}{(6)(4)(5)} ((15)^2 + (15)^2 + (15)^2 + (15)^2) \right] - (3)(6)(5)$$

$$\text{i.e.} \quad = \frac{3}{30} (900) - 90 = 0$$